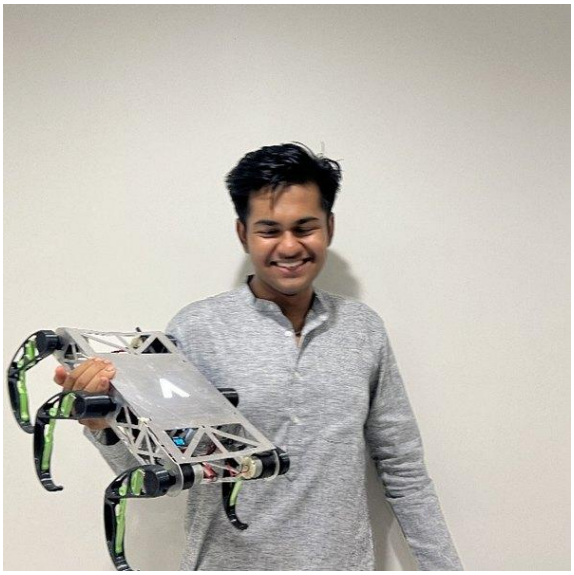
Aditya Dutta

Robotics | Embedded Systems
Power Electronics | Leadership

Bengaluru, India
+91 98744 58859
adityadutta2006@gmail.com
aditya.mitblr2024@learner.manipal.edu
thedutta.github.io github.com/thedutta
linkedin.com/in/aditya-dutta-058808213



Skills

Embedded & Firmware Embedded C/C++ · STM32 (G4, F4) and ESP32 · bare-metal real-time firmware · low-power design · SPI, I^2C, UART, CAN-FD · hardware debugging.

Power Electronics & PCB Multi-layer PCB design (Altium) · three-phase BLDC drive · gate drivers and FET selection · Li-ion charging and battery management · switching regulators and LDO rails · current sensing · signal integrity · design for manufacture.

Control & Robotics Field-oriented control · closed-loop position, velocity and torque · encoder feedback · sensor fusion · distributed Jetson + MCU architectures · ROS 2 · Isaac Sim.

Mechanical & Fabrication Fusion 360 · mechanism design, bio-inspired and functional · 3D printing · CNC and laser fabrication · manual PCBA · rapid iteration under constraint.

Leadership & Technical Direction System architecture planning · design reviews · subsystem decomposition · mentoring university engineering teams.

Information

Undergraduate, Computer Science & Engineering

Manipal Institute of Technology, Bengaluru (MIT-B)

Third year · expected graduation 2028

Academic interests Robotics, embedded systems, autonomous platforms, product design and manufacturing.

Current roles President, RADAR Student Robotics Club. Co-Founder & Technical Lead, Sylvara Inc.

Hometown Kolkata, West Bengal

Tools & Software

Autodesk Fusion 360 · Altium Designer · ROS 2 Jazzy · NVIDIA Isaac Sim · Jetson compute · MATLAB · Oracle SQL · Adobe Photoshop · BambuStudio & Cura · Linux terminal

Multi-layer PCB design · Advanced CAD · 3D printing · CNC & laser manufacturing · Manual PCBA · Rapid prototyping · Product conceptualisation

ONGOING RESEARCH

CatBot

BIOMIMETIC QUADRUPED · DISTRIBUTED EDGE NODE

OBJECTIVE

Build a quadruped held to the physical envelope of the animal it copies — the footprint and all-up mass of a wildcat, under 5 kg — and give it enough onboard autonomy to work unsupervised where a wheeled platform cannot go.

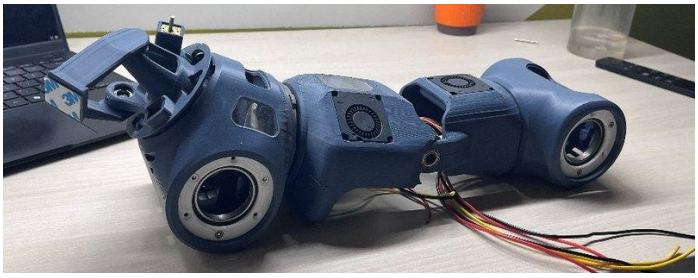
SYSTEM OVERVIEW

CatBot is a 19-degree-of-freedom (21 with the optional tail), 4.8 kg biomimetic quadruped built around a decentralised electronic nervous system. The obvious way to build a quadruped is one controller wired to every motor; that stops working the moment mass, latency and wiring matter. CatBot inverts it — every actuator carries its own FlexNode closing a tens-of-kilohertz current loop millimetres from the windings, and what runs down the body is a CAN-FD pair and power.

Above that sit two tiers. A Jetson Orin Nano Super does perception and planning — stereo depth, visual-inertial SLAM, local reasoning. It is non-real-time by design and allowed to crash. Beneath it an STM32 brainstem holds reflexes, power sequencing and watchdogs, and stays alive when the Jetson does not. Sensing is strictly passive, so several units can share a space without blinding one another.



The complete assembly — mechanical design closed, URDF export next.



Printed hip and limb actuator assembly.

VITALS

MASS	FREEDOM	PEAK POWER	BATTERY	MAX JUMP	SPINE
4.8 kg	**19 + 2 DoF**	**2.5 kW**	**266 Wh**	**1.4 m**	**CAN-FD 5 Mbit**

KEY ENGINEERING DECISIONS

Distributed over central control

One node per actuator. Adding a limb is adding nodes, not rewiring the robot, and splitting the chain left/right means one bus fault takes down one side, not the machine.

Passive-only perception

Depth from parallax and scene texture, not a projected dot pattern. Two CatBots can work the same room, and neither announces itself in infrared.

Knees driven by belt

Six knee motors moved up off the distal limb through a 14:1 dual-stage reduction. Limb inertia is what a leg must accelerate every stride, and no controller tuning recovers what the mechanism gives away.

A genuine zero-watt state

The Jetson and every motor controller power down completely while the STM32 draws microamps watching an analogue microphone line. A specific acoustic signature whips the head onto the sound and boots the GPU behind it.

TECHNICAL SPECIFICATIONS

Compute Jetson Orin Nano Super for perception and planning; custom STM32 brainstem for reflexes, power and watchdogs; one STM32G4 FlexNode per actuator.

Actuation 19 DoF, plus 2 optional in the tail — 4 × GIM8108-8 quasi-direct-drive hips, 6 × 5010 360 kV knees on 14:1 S3M belts, 2 × gimbal head motors, 7 × DS3235/DS3230 servos.

Sensing OAK-D Lite passive stereo, distributed VL53L5CX time-of-flight, Knowles MEMS mics into the ultrasonic band, per-node IMU and magnetic encoders.

Power 8S2P Molicel INR21700-P45B, ~29.6 V nominal.

Structure Forged carbon fibre and titanium as production intent; 3D-printed structure and CNC limb mounts in hand.

Status Design closed, torso assembled; FlexNode v1.0 running closed-loop FOC on the test stand (Oct 2026). CAN multi-node integration is the current work.

ONGOING RESEARCH

FlexNode

DISTRIBUTED CAN-FD COMPUTE NODE

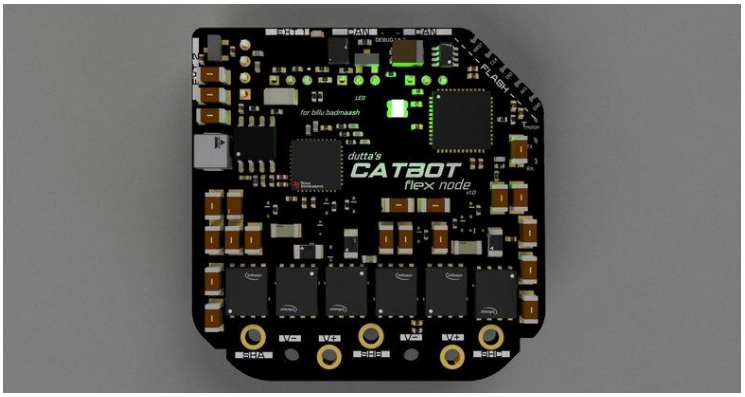
OBJECTIVE

Collapse mobile-robot electronics down to one repeatable part, so that any sensor-fused robot reduces to buying FlexNodes, actuators and sensors, wiring power and CAN, and developing no electronics at all.

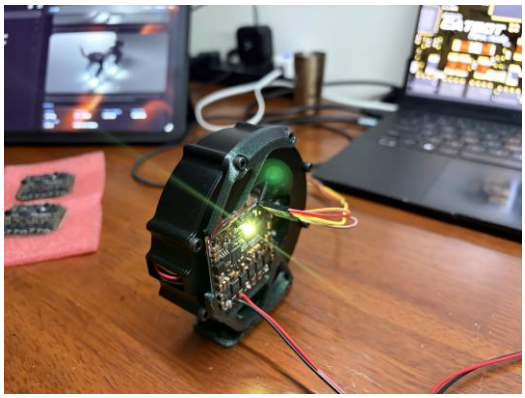
SYSTEM OVERVIEW

FlexNode fuses a high-current field-oriented motor controller and a sensor-fusion front end onto one compact PCB hanging off a CAN-FD bus. It began as a fork of the open-source mjbots moteus r4.11 — proven power stage, DRV8353S gate driver and STM32G4 core kept intact — and adds what a distributed node needs: inertial sensing, optional ranging, servo/aux and I²C ports, and its own logic supply.

The fleet runs one firmware image: node identity is configuration, not code, so thirteen joints means thirteen identical boards, two wires and a config file each. The first article booted on 6 September 2026, turned a motor two days later, and closed the loop in October — full field-oriented control on its own encoder.



FlexNode v1.0 — STM32G4, DRV8353S and the six-FET power stage.



Closed-loop FOC on the test stand — October 2026.

POWER STAGE

INPUT	PEAK PHASE	CONTINUOUS	PEAK POWER	PWM	CONTROL LOOP
10–44 V	**100 A**	**12 / 32 A**	**900 W @ 30 V**	**15–60 kHz**	**15–30 kHz**

TI DRV8353S into six 60 V / 1.6 mΩ N-channel MOSFETs with per-phase current sensing; continuous current quoted without and with thermal management. Ratings inherited from r4.11 by keeping its topology, gate driver and FET selection; FlexNode's stage is electrically and thermally validated, with full-rated-load characterisation to come.

KEY ENGINEERING DECISIONS

One image, N configurations

A reserved register block lets a node advertise the peripherals it carries, so the host discovers the fleet rather than being told about it.

Deleted the autodetection

Upstream probes strap pins at boot to identify the board; FlexNode repurposes those pins, so identity is hardcoded and the probe deleted — ~1.7 KiB of flash returned.

Spun before trusted

Open-loop voltage FOC from a current-limited 13 V / 1 A supply, behind software aborts on bus sag, overcurrent and FET temperature: eight loops, ~140 s, zero faults.

A two-line fix instead of a respin

The v1.0 copper swaps two motor phases against the reference. Rather than rework every board, the pin map is swapped in the hardware-description layer and the FOC code stays untouched.

TECHNICAL SPECIFICATIONS

Microcontroller STM32G473CEU6 — Cortex-M4F at 170 MHz, 512 KB flash, UFQFPN-48.

Feedback AS5047P on-axis magnetic encoder over SPI.

Onboard sensing LSM6DS3TR-C 6-axis IMU; optional VL53L7CX 8×8 time-of-flight.

Communications CAN-FD via TCAN334G at 1 Mbit / 5 Mbit, daisy-chained, 120 Ω termination on leaf nodes.

Logic rails LGS5160C synchronous buck to 5 V / 3 A, ME6216A33 LDO to 3.3 V.

Protocol Upstream register protocol plus a FlexNode block at 0x080–0x0FF for capability discovery.

Lineage Built on mjbots moteus r4.11 (Apache-2.0).

Status Booted 6 Sept; spun 8 Sept; closed-loop FOC running Oct 2026. CAN fleet next.

ONGOING STARTUP · CO-FOUNDER & TECHNICAL LEAD

Sylvara™

ULTRA-LOW-POWER MODULAR PLANT-CARE ECOSYSTEM

OBJECTIVE

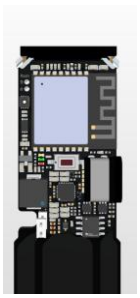
A compact, modular plant-care system for year-scale autonomous operation, retrofittable into existing gardens and farms, under extreme power, size and cost constraints.

SYSTEM OVERVIEW

Sylvara is a production-oriented IoT platform built for long-term unattended deployment, integrating environmental sensing, soil monitoring, energy harvesting and wireless connectivity into a very compact modular form factor.

Incorporated in Canada as Sylvara Inc. (OCN 1001215100, 23 April 2025), trademarked and registered, funded with CAD 10,000 in summer 2025, and supported by the Morrissette Institute for Entrepreneurship with active venture interest. The first production batch — 100 units of v1.3 — is in manufacture.



Left to right — rendered in singular plant use: solar harvesting, no wiring, no scheduled maintenance · v1.2 board in hand, beside its CAD model; prototype batch of 10, fully hardware validated · board in enclosure · PCB top view of the electronics region, 22 × 42 mm.

VITALS

AVG DRAW	DARK RUNTIME	DAILY SUN	PCB	RADIO	BATCH
53 µA	**1.3 years**	**30 min**	**22 × 42 mm**	**Wi-Fi + BLE**	**100 · v1.3**

TECHNICAL SPECIFICATIONS

Microcontroller ESP32-C3-WROOM-02 — Wi-Fi, BLE, RISC-V.

Power system TI BQ25798 Li-ion management, solar harvesting, USB-C PD charging.

PCB 22 × 42 mm, multi-layer integrated design.

Sensing Temperature, humidity, light, soil moisture and capacitive water-level monitoring.

Status v1.3 released to production; first batch of 100 units in manufacture.

KEY ENGINEERING DECISIONS

Energy-first architecture

Power budgeting shaped component selection, firmware scheduling and communication strategy rather than being optimised afterwards — near-perpetual operation follows from roughly twenty minutes of daily sunlight, not a larger battery.

High-density integration

Sensing, power management, charging and compute on one 22 × 42 mm PCB through careful IC selection, without giving up reliability or serviceability.

Design for production

Built for scalable manufacturability from the outset — optimised part dependencies, waterproof enclosure integration, and a BOM that survives a real assembler.

REVISION HISTORY

Four revisions and what each taught — the failures are where the engineering actually happened.

v1.0 Dual-PCB architecture abandoned on cost and assembly complexity; hobby-grade TP4056 charging IC dropped for an industry-grade TI part on thermal, space and telemetry grounds.

v1.1 BQ25798 misconfigured and bypassed to test; GPIO capacitive sensing non-linear and noisy, replaced with a TLC555 timer circuit; 10 µF 0402 capacitors a sourcing and performance problem, moved to 1206.

v1.2 EMI modelled incorrectly in MATLAB but measured within spec on the bench; ESD/TVS diodes missed on the USB data lines; the USB-C IPX9 gasket did not fit, from a misread tolerance.

v1.3 Gasket tolerance corrected; BQ25798 NTC battery cutoff restored. In production.

COMPLETED PROJECT

RoVolt

WHEG ROBOT & MOBILE AGRICULTURAL PLATFORM

OBJECTIVE

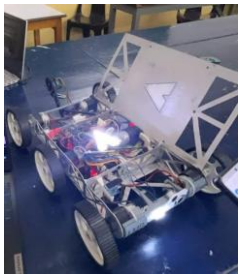
Build a terrain-adaptive robotic platform capable of reliable locomotion across uneven farmland, then extend it into a smart agricultural system supporting soil interaction, sensing and basic farming operations.

SYSTEM OVERVIEW

RoVolt began as a Class 11 project built around a WHEG — wheel-leg hybrid — locomotion system. The platform was fully CAD-modelled and fabricated from laser-cut aluminium, manually brazed into a rigid but serviceable chassis. Locomotion came from custom 3D-printed WHEGs on a push-rod suspension, driven by DC motors with encoder feedback under a Raspberry Pi.

It demonstrated stable traversal across varied terrain including loose soil. For competitions the system was reconfigured from the original 2×3 WHEG layout into a three-wheel configuration and fitted with a rear-mounted arm carrying soil-moisture sensing, ploughing and seeding attachments. An IMU added directional awareness while the existing encoders were reused for motion feedback, and a companion mobile application enabled real-time monitoring. I led hardware integration and low-level firmware. The platform was shown at multiple expos and competitive hackathons, taking top placements and awards, and was covered in the press.







Left to right — laser-cut aluminium, brazed by hand · assembled chassis, electronics bay open · press coverage, Hackathon '23.



On the expo table — the platform, its CAD, and the companion app.

TECHNICAL SPECIFICATIONS

Microprocessor Raspberry Pi 3 on a proto-board.

Control TB6612FNG dual motor drivers with quadrature encoders.

Locomotion Custom 3D-printed WHEGs on a push-rod suspension.

Sensors Wheel encoders, IMU, soil-moisture sensor, arm kinematics.

Fabrication Laser-cut aluminium chassis, brazed by hand; 3D-printed mounts and WHEGs.

KEY ENGINEERING DECISIONS

WHEG-based locomotion

A wheel-leg hybrid balances mechanical simplicity against terrain adaptability far better than a conventional wheeled platform.

Hands-on fabrication

Designing, laser-cutting and brazing the chassis myself kept structural integrity and iteration speed in my own hands rather than a supplier's.

Platform adaptability

Reconfiguring the locomotion layout and adding a modular agricultural arm demonstrated versatility beyond mobility alone.

Sensor reuse and expansion

Existing encoders were reused while inertial sensing was added — better directional stability, no extra complexity.

PRESIDENT · MIT-B STUDENT ROBOTICS CLUB

Mentorship

LEADERSHIP & TECHNICAL DIRECTION

As President of RADAR, one of the most active technical clubs at Manipal Institute of Technology Bengaluru, I mentor multiple student teams, translating ambitious ideas into technically feasible systems against real skill and resource constraints. The role is architectural guidance, design reviews, subsystem decomposition, and helping teams close the gap between concept and execution.

ONGOING MENTORED PROJECTS

TrackMate An autonomous athletic-track pacer focused on consistent velocity control and lap accuracy. In collaboration with Tenvik Sports.

Arachne A spider-robot reconnaissance platform exploring bio-mimetic locomotion and terrain adaptability — a scaled-down mirror of CatBot.

BattleRobot team Competitive combat robotics: high-current power systems, mechanical durability and failure-resilient design.

CO-DESIGNED & COMPLETED

DoorThing — retrofit electronic door unlock
A low-cost, ID-card-based door unlock retrofit built around reliability, power efficiency and ease of installation — embedded control and secure actuation under real institutional constraints.

RC vehicle platform — event robotics kit
An iterable, modular RC platform for a student robotics event, robust enough to be repurposed for the BattleRobot at short notice. Built on Arduino and NRF modules around a compact interconnection PCB, it became the teaching entry point for drivetrain fundamentals and motor control.

CSEA Academic Hub
A student-facing hub for CSE Section A, grouping tools, materials and presentations in one place.



Club orientation day, 2025.



TrackMate on the track.



BattleRobot drivetrain on the bench.





RC vehicle platform — the drawing sheet, and at the club event.

OUTSIDE ENGINEERING

I maintain long-distance endurance training and regular exercise as a way of sustaining discipline, recovery and long-term cognitive performance. I currently run with the track team, and placed third in a recent half marathon.



Conclusion

ENGINEERING TRAJECTORY & INTENT

My work in robotics follows a lineage rather than a collection of independent projects. RoVolt was the origin point: a WHEG-based mobile robot that demonstrated reliable locomotion across uneven terrain and was later adapted into a functional agricultural platform with sensing, actuation and field interaction. Building it end to end — mechanically fabricated, electrically integrated, iteratively modified — established my grounding in how robots physically engage with real environments, and its results across expos and hackathons validated both the platform and a systems-level way of working.

From RoVolt came two deliberate and divergent evolutions. Sylvara is the agricultural branch, shifting the core challenge from mobility to endurance, integration and scalability inside a funded startup: energy-first thinking, ultra-low-power electronics, dense PCB integration and production-oriented design. CatBot is the autonomy branch, pushing toward agile locomotion, perception-guided decision-making and real-time control under strict size, weight and reliability constraints.

FlexNode is what happened when CatBot's hardest constraint was taken seriously enough to become its own product. Nineteen joints could not each be wired back to one controller, so the electronics were pushed out to the joints and made identical and configurable. It is the piece I would hand to someone else: buy the nodes, wire power and CAN, and skip the year of electronics development that otherwise stands between an idea for a robot and a robot.

Together these reflect a progression in both technical scope and engineering responsibility — from a mobile robotic platform, to a commercial-grade embedded system, to a research-driven autonomous robot, to a reusable hardware product. The through line is a preference for deployable systems, constraint-aware design and holistic integration over isolated technical novelty.

Moving forward, my goal is to keep building autonomous robotic systems that extend human capability in environments where risk, scale or endurance limit direct human presence.

Documentation for every project above is published at **thedutta.github.io**.

PROJECT INDEX

CatBot Biomimetic quadruped, 19 + 2 DoF, 4.8 kg, CAN-FD spine.
thedutta.github.io/projects/catbot

FlexNode Distributed CAN-FD compute node. Built on moteus.
thedutta.github.io/projects/flexnode

Sylvara Ultra-low-power plant care. Incorporated, in production.
sylvara.life

RoVolt WHEG robot and mobile agricultural platform.
thedutta.github.io/projects/rovolt

RADAR TrackMate, Arachne, BattleRobot, DoorThing, RC platform.
thedutta.github.io/projects

Notes Engineering write-ups and process notes.
thedutta.github.io/blog